

# MagMat-Discovery

Physics-Informed Machine Learning and Screening for Magnetic Materials

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## System Architecture, Methodology & Model Validation

Technical Report & Model Guide

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## Executive Summary

Permanent magnetic materials with high energy density and thermal stability are essential for electrification, renewable energy generation, and precision industrial systems. However, computational discovery of novel magnetic compounds is hindered by the fundamental disconnect between chemical composition, crystal symmetry, and microstructural coercivity mechanisms. While intrinsic magnetic properties such as Curie temperature ( $T_C$ ) and Néel temperature ( $T_N$ ) depend primarily on local chemical bonding and exchange coupling, extrinsic properties such as coercivity ( $H_c$ ) and maximum energy product ( $(BH)_{\max}$ ) are strongly dictated by crystal anisotropy, grain morphology, and domain wall pinning.

**MagMat-Discovery** is an open machine learning and materials screening platform designed to address these multi-scale challenges. The system combines:

- **Curated Experimental & Crystallographic Data:** 51,394 experimental records from the Non-Equilibrium Magnetic Materials Database (NEMAD), cross-referenced with DFT crystal symmetries (space groups 1–230, point groups, and theoretical densities) from the Materials Project funded by the U.S. Department of Energy (DOE).
- **Pretrained Foundation Models:** Equivariant atomic representations from MACE-MP-0 and 3D periodic crystal synthesis with thermodynamic convex hull distance ( $E_{\text{hull}}$ ) from the DAO diffusion crystal engine.
- **In-House Domain Feature Engineering:** Multi-tier descriptors encoding Slater-Pauling valence filling, Goodenough-Kanamori superexchange proxies, spin-orbit scaling, and Stevens crystal-field operator terms.
- **In-House Graph Transformer: MagFormer,** a custom 384-dimensional self-attention graph transformer trained from scratch to embed chemical stoichiometry without requiring 3D coordinates.
- **Micromagnetic Physical Cascades:** Cascaded model coupling linking intrinsic anisotropy and magnetization predictions to extrinsic coercivity via the Kronmüller relationship, while strictly bounding energy products by the thermodynamic ceiling ( $(BH)_{\max} \leq \frac{1}{4}\mu_0 M_s^2$ ).
- **Calibrated Uncertainty Ranges:** Conformal prediction intervals constructed from out-of-fold residuals, guaranteeing distribution-free 90% coverage for screening decisions.

Evaluated under leak-free 5-fold grouped cross-validation, the platform demonstrates strong predictive capability on ground-state magnetic phase (91.9% accuracy) and transition temperatures ( $R^2 = 0.82$  for  $T_C$ ,  $R^2 = 0.74$  for  $T_N$ ). For extrinsic hysteresis properties, the models achieve moderate predictive accuracy ( $R^2 = 0.46$ – $0.62$ ), providing a fast, physically grounded ranking tool to prioritize candidate magnets before costly laboratory synthesis.

**Keywords:** Magnetic Materials, Physics-Informed Machine Learning, Graph Transformers, Conformal Calibration, Coercivity, Permanent Magnets, Materials Screening.

Table 1: Glossary of magnetic properties, physical units, and computational terms.

| Term                                  | Typical Unit      | Physical Meaning / Definition                                                                                                              |
|---------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| $T_C$ (Curie Temp.)                   | K                 | Temperature above which a ferromagnetic material loses its spontaneous magnetization.                                                      |
| $T_N$ (Néel Temp.)                    | K                 | Temperature above which an antiferromagnetic material becomes paramagnetic.                                                                |
| $M_s$ (Saturation Mag.)               | emu/g or T        | Maximum magnetization achieved in an applied magnetic field. Reported in mass units (emu/g) or magnetic polarization $\mu_0 M_s$ (T).      |
| $H_c$ (Coercivity)                    | A/m or kA/m       | Applied magnetic field required to reduce magnetization back to zero after saturation.                                                     |
| $K_1$ (Anisotropy Constant)           | J/m <sup>3</sup>  | Energy density required to rotate magnetization away from the preferred crystal axis.                                                      |
| $(BH)_{\max}$ (Energy Product)        | kJ/m <sup>3</sup> | Primary figure of merit for permanent magnets, representing stored magnetic energy density.                                                |
| $\theta_p$ (Paramagnetic Curie Temp.) | K                 | High-temperature intercept from Curie-Weiss susceptibility fitting; its sign reflects dominant exchange coupling.                          |
| FM / AFM / NM                         | Discrete          | Ground-state magnetic ordering: Ferromagnetic (parallel), Antiferromagnetic (antiparallel), or Non-magnetic.                               |
| Out-of-Fold (OOF)                     | –                 | Predictions made on data held out during cross-validation, ensuring unbiased inputs to downstream models.                                  |
| Conformal Range                       | Physical          | A calibrated uncertainty interval constructed from validation residuals, expected to contain the true value approximately 90% of the time. |
| dex                                   | log <sub>10</sub> | Decimal exponent (order of magnitude). An error of 0.65 dex means a factor of $10^{0.65} \approx 4.5$ .                                    |

## Glossary of Key Terms

To assist readers from different materials science and computational backgrounds, Table 1 summarizes the primary magnetic properties and machine learning terminology used throughout this report.

## 1 Project Background and Objectives

Discovering high-performance permanent magnets with reduced rare-earth content is important for electric vehicles, wind turbines, and industrial machinery. The strongest commercial magnets rely heavily on critical rare-earth elements, primarily Neodymium (Nd), Dysprosium (Dy), and Samarium (Sm), which are vulnerable to supply disruptions, volatile pricing, and environmentally destructive extraction methods. Developing sustainable alternative magnets requires compounds that satisfy three strict physical requirements: elevated Curie temperature ( $T_C > 400$  K) to resist thermal demagnetization under operating conditions, high saturation magnetization ( $\mu_0 M_s > 1.0$  T) to deliver strong magnetic flux density, and substantial coercivity ( $H_c > 800$  kA/m) to withstand opposing demagnetizing fields.

Applying machine learning to accelerate this discovery faces three major scientific hurdles:

1. **Disparate Physical Scales:** Target properties operate on fundamentally different length scales. Intrinsic properties like  $T_C$  and  $T_N$  correlate well with chemical composition because they are governed by atomic exchange coupling. Extrinsic properties like  $H_c$  and  $(BH)_{\max}$  depend critically on crystal anisotropy, grain morphology, boundary phases, and defect pinning, causing composition-only models to struggle.
2. **Data Leakage across Polymorphs and Dopants:** Experimental materials databases frequently contain numerous entries of the same parent alloy with minor dopant variations or different measurement temperatures. Standard random data splitting places near-identical stoichiometries into both training and validation sets, producing artificially high validation scores that fail to generalize to novel chemical systems.
3. **Violation of Thermodynamic and Physical Bounds:** Independent machine learning models predict each property in isolation, which frequently yields physically impossible combinations—such as predicted energy products exceeding the theoretical limit ( $\frac{1}{4}\mu_0 M_s^2$ ) or finite coercivity predicted for non-magnetic compositions.

MagMat-Discovery resolves these challenges by combining leak-free grouped cross-validation, multi-tier physics descriptors, a custom stoichiometric graph transformer (**MagFormer**), micromagnetic constraint cascades, and distribution-free conformal calibration into an integrated discovery engine.

## 2 Data Curation and Split Design

### 2.1 Experimental and Crystallographic Data Sources

The platform aggregates experimental measurements with crystallographic symmetry databases to establish a reliable foundation for model training and evaluation:

- **NEMAD Experimental Ground Truth:** We curate 51,394 experimental measurements from the Non-Equilibrium Magnetic Materials Database [1]. These records supply experimental target values across 8 properties: ground-state magnetic phase, Curie temperature ( $T_C$ ), Néel temperature ( $T_N$ ), saturation magnetization ( $M_s$ ), magnetocrystalline anisotropy ( $K_1$ ), coercivity ( $H_c$ ), maximum energy product ( $(BH)_{\max}$ ), and paramagnetic Curie temperature ( $\theta_p$ ).
- **Materials Project (U.S. Department of Energy / DOE):** From the open DFT repository developed under the U.S. DOE Office of Science [2], we extract structural symmetries: space group numbers (1–230), Hermann-Mauguin symbols, crystal systems, point groups, and theoretical mass densities. To ensure models remain applicable to uncharacterized materials

and to prevent target leakage, all DFT-calculated total energies, formation energies, and computed ground-state magnetic moments are strictly excluded from the input feature set.

- **Bilbao MAGNDATA:** Magnetic space group ground truths and Shubnikov symmetry classifications are incorporated for magnetic structure validation.

## 2.2 Data Cleaning and Grouped Cross-Validation Partitioning

Raw experimental records are sanitized by canonicalizing chemical formulas to coprime integer stoichiometries using `pymatgen`. Properties spanning multiple orders of magnitude ( $H_c, K_1, (BH)_{\max}$ ) are transformed using base-10 logarithmic or symmetric-logarithmic scales to stabilize variance during training.

To ensure realistic performance estimates, cross-validation is performed using a 5-fold `GroupKFold` protocol partitioned strictly by `reduced_formula`. Under this partitioning, identical chemical compositions and pure polymorphs (such as different crystal modifications of  $\text{Fe}_3\text{Pt}$  or multiple measurements of  $\text{Nd}_2\text{Fe}_{14}\text{B}$ ) are isolated entirely within a single fold. This guarantees that test folds contain only unseen compounds, evaluating whether the models can generalize across chemical families rather than memorizing known parent stoichiometries.

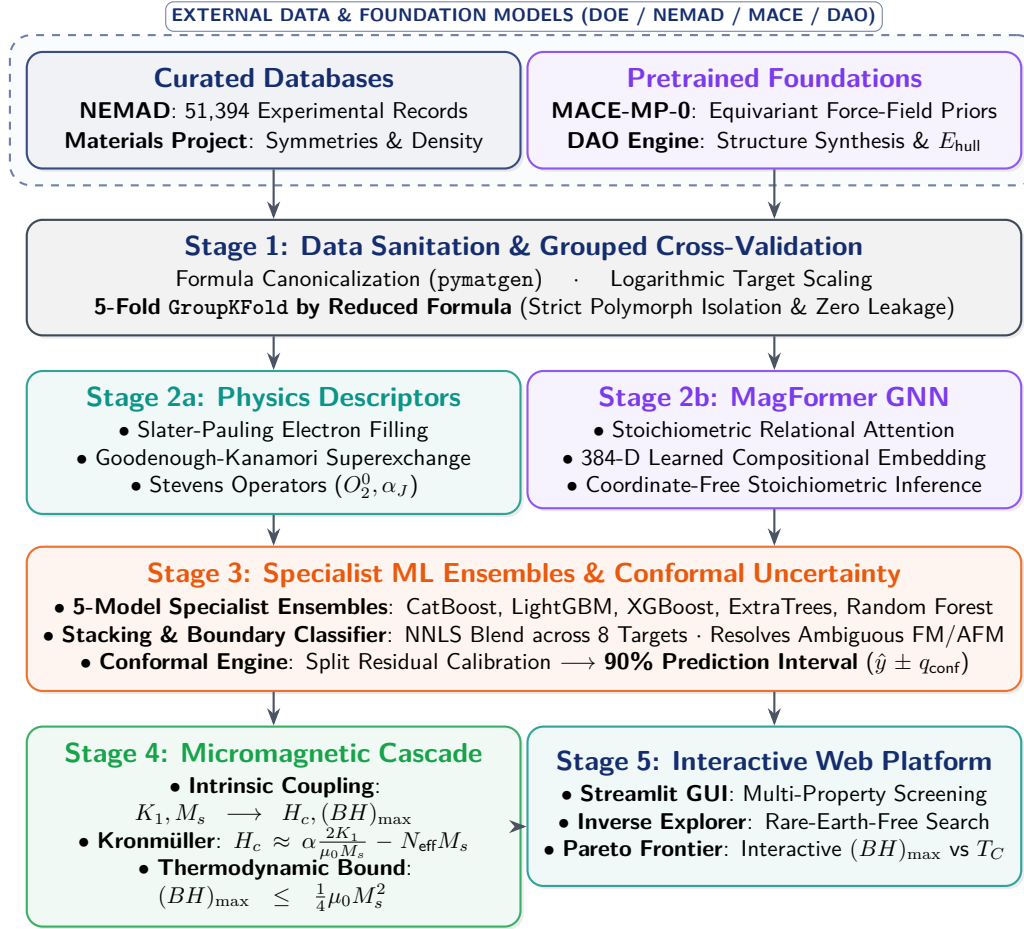


Figure 1: System architecture and end-to-end data flow of MagMat-Discovery. Curated databases (NEMAD, Materials Project funded by DOE) and pretrained foundation models (MACE-MP-0, DAO) provide baseline ground truths and structural embeddings. In-house pipeline components (MagFormer stoichiometric transformer, multi-tier physics featurizers, 5-model specialist ensembles, micromagnetic physical cascades, and conformal uncertainty calibration) perform leak-free training and screening.

### 3 Methodology and System Architecture

#### 3.1 System Architecture and Foundation Provenance

Figure 1 illustrates the overall architecture of MagMat-Discovery. A guiding design requirement of the platform is clear separation of model provenance: distinguishing what is imported from external foundation models and government databases from what was designed, trained, and validated in-house.

Table 2 provides a detailed accounting of data and model origins across all pipeline stages. External foundation resources provide experimental ground truths, crystal symmetries, equivariant force-field embeddings, and generative 3D crystal structures. All property regression models, boundary classifiers, micromagnetic cascades, and conformal uncertainty calibrators were formulated, trained, and validated in-house.

Table 2: Provenance of databases, foundation models, and in-house algorithmic components in MagMat-Discovery.

| Component                             | Category                                     | Origin / Training Source                                                        | Exact Role in MagMat-Discovery                                                                                                                                                                       |
|---------------------------------------|----------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Materials Project (U.S. DOE)</b>   | External DFT Database                        | High-throughput DFT funded by U.S. DOE Office of Science [2]                    | Supplies crystallographic symmetries (space groups 1–230), point groups, crystal systems, and densities. DFT total energies and magnetic moments are excluded to prevent target leakage.             |
| <b>NEMAD Database</b>                 | External Experimental Database               | Curated experimental literature compilation (51,394 entries) [1]                | Provides ground-truth experimental targets ( $T_C, T_N, M_s, H_c, K_1, (BH)_{\max}$ , phase) for 5-fold grouped model training and benchmark validation.                                             |
| <b>MACE-MP-0 Foundation GNN</b>       | External Pretrained Foundation Model         | Equivariant neural network pretrained on DOE MP trajectories [3]                | Extracts higher-order equivariant atomic cluster expansion descriptors, cohesive energies, and stress tensors for compounds with known 3D crystal structures.                                        |
| <b>DAO Crystal Foundation Engine</b>  | External Pretrained Foundation Model         | Diffusion crystal foundation model (Huang et al., <i>Nat. Commun.</i> 2026) [4] | Employs DAO-G to synthesize 3D periodic crystal structures from stoichiometry and DAO-P to calculate thermodynamic distance to convex hull ( $E_{\text{hull}}$ in eV/atom).                          |
| <b>MagFormer Stoichiometric GNN</b>   | In-House Architecture (Trained from Scratch) | Designed and trained in this pipeline on chemical compositions                  | Stoichiometric self-attention graph transformer that embeds chemical formulas into 384-D representation vectors without requiring atomic coordinates.                                                |
| <b>Multi-Tier Physics Descriptors</b> | In-House Domain Feature Engineering          | Derived from solid-state and exchange physics theory                            | Handcrafted features encoding Slater-Pauling valence filling ( $\text{VEC} \approx 8.3$ ), Goodenough-Kanamori superexchange proxies, spin-orbit terms, and Stevens operators ( $O_2^0, \alpha_J$ ). |
| <b>5-Model Specialist Ensembles</b>   | In-House Machine Learning Models             | Trained on leak-free 5-fold grouped splits in this work                         | Diverse ensemble (CatBoost, LightGBM, XGBoost, ExtraTrees, Random Forest) with non-negative least squares meta-blending for all 8 target properties.                                                 |
| <b>Boundary Phase Classifier</b>      | In-House Machine Learning Model              | Trained on ambiguous boundary compounds in this work                            | Secondary classifier resolving close ferromagnetic vs. antiferromagnetic ground states ( $ P(\text{FM}) - P(\text{AFM})  < 0.20$ ).                                                                  |
| <b>Micromagnetic Physical Cascade</b> | In-House Physical Post-Processing            | Formulated from micromagnetic theory [5]                                        | Enforces the Kronmüller coercivity relationship and caps maximum energy product at the thermodynamic limit $(BH)_{\max} \leq \frac{1}{4}\mu_0 M_s^2$ .                                               |
| <b>Conformal Uncertainty Engine</b>   | In-House Uncertainty Calibration             | Split conformal prediction on out-of-fold residuals [6]                         | Computes rigorous distribution-free 90% uncertainty intervals ( $y \pm q_{\text{conf}}$ ) calibrated specifically to out-of-fold validation errors.                                                  |

### 3.2 Dual Representation Learning and Structure Synthesis

To capture both fundamental chemistry and crystal physics, the pipeline employs a dual representation approach:

**1. Multi-Tier Physics Descriptors:** Input compositions are featurized through four hierarchical physical tiers (detailed mathematically in Appendix A):

- *Tier 1 (Stoichiometry and Elements):* Elemental mass fractions, atomic numbers, covalent radii, and Pauling electronegativities.
- *Tier 2 (Electronic Structure and Superexchange):* Valence electron concentration (VEC) referenced to the Slater-Pauling maximum ( $\text{VEC} \approx 8.3$ ), Goodenough-Kanamori superexchange proxies capturing anion-mediated exchange, and spin-orbit proxies ( $\propto Z^4$ ) for heavy-element interactions.
- *Tier 3 (Crystal Field and Anisotropy):* Crystal system indicators and second-order Stevens operator terms ( $O_2^0, \alpha_J$ ) derived from Materials Project symmetries, indicating easy-axis ( $K_1 > 0$ ) versus easy-plane preferences.
- *Tier 4 (Equivariant Force-Field Embeddings):* High-order equivariant atomic representations extracted from MACE-MP-0 [3] for compounds with known crystal coordinates.

**2. MagFormer Stoichiometric Graph Transformer:** When screening uncharacterized or hypothetical alloys, 3D atomic coordinates are unavailable. To learn compositional representations directly from formula strings, we developed **MagFormer**, a custom stoichiometric graph transformer trained from scratch. MagFormer treats a chemical formula as a fully connected element graph where nodes are initialized with elemental properties and multi-head attention is biased by pairwise chemical differences ( $|\mathbf{p}_i - \mathbf{p}_j|$ ). Tri-part stoichiometry-weighted pooling yields a dense 384-dimensional representation vector that is passed to downstream tree models (Appendix B).

**3. DAO Crystal Synthesis and Synthesizability Screening:** When 3D structural validation is required for a new stoichiometry, the platform invokes the **DAO** diffusion foundation engine [4]. The DAO-G generator synthesizes periodic crystal unit cells (lattice vectors and atomic coordinates) directly from chemical formulas. Symmetries are identified with **pymatgen**, while the DAO-P module evaluates the energy above the convex hull ( $E_{\text{hull}}$ ). Compositions with  $E_{\text{hull}} < 0.05$  eV/atom are flagged as readily synthesizable, ensuring hypothetical screening candidates are crystallographically viable.

### 3.3 Specialist Ensembles, Physical Cascades, and Uncertainty Calibration

For each of the 8 target properties, predictions are generated by an ensemble of five diverse regressors: CatBoost, LightGBM, XGBoost, ExtraTrees, and Random Forest. Predictions are synthesized via non-negative least squares (NNLS) meta-blending, which optimizes ensemble weights on out-of-fold predictions. A dedicated secondary boundary classifier resolves ambiguous phase candidates where predicted FM and AFM probabilities differ by less than 0.20.

To enforce physical laws across intrinsic and extrinsic targets, the pipeline executes a micromagnetic physical cascade:

1. Intrinsic properties ( $K_1$  and  $M_s$ ) are predicted first.
2. Out-of-fold estimates of  $K_1$  and  $M_s$  are supplied as inputs to the coercivity ( $H_c$ ) and energy product ( $(BH)_{\text{max}}$ ) regressors. This grounds coercivity predictions in the Kronmüller micromagnetic relation [5]:

$$H_c \approx \alpha_{\text{micro}} \frac{2K_1}{\mu_0 M_s} - N_{\text{eff}} M_s \quad (1)$$

where  $\alpha_{\text{micro}}$  accounts for microstructural reduction (grain boundary defects) and  $N_{\text{eff}}$  is the effective demagnetizing factor.

3. Predicted energy products are capped at the thermodynamic limit:  $(BH)_{\text{max}} \leq \frac{1}{4} \mu_0 M_s^2$ .

Every continuous prediction is accompanied by a distribution-free uncertainty range using split conformal prediction [6]. The 90th percentile of absolute out-of-fold validation residuals establishes a calibration margin:

$$q_{\text{conf}} = \text{Quantile}_{0.90} (\{|y_i - \hat{y}_i|\}_{i=1}^n) \quad (2)$$

The resulting prediction interval  $[\hat{y} - q_{\text{conf}}, \hat{y} + q_{\text{conf}}]$  provides rigorous 90% coverage without requiring parametric error assumptions.

Table 3: Comprehensive 5-fold GroupKFold cross-validation results across all 8 magnetic targets.

| # | Target Property                         | Unit              | Scale             | Count $n$ | Primary Metric        | Mean Error          | Predictive Utility                            |
|---|-----------------------------------------|-------------------|-------------------|-----------|-----------------------|---------------------|-----------------------------------------------|
| 1 | Ground-State Phase                      | Discrete          | Class             | 35,045    | <b>91.9% Accuracy</b> | Macro $F_1$ : 0.908 | <b>Strong</b> (Reliable phase filtering)      |
| 2 | Curie Temperature ( $T_C$ )             | K                 | Linear            | 15,518    | $R^2 = 0.816$         | MAE = 69.8 K        | <b>Strong</b> (Thermal threshold filter)      |
| 3 | Néel Temperature ( $T_N$ )              | K                 | Linear            | 7,734     | $R^2 = 0.741$         | MAE = 48.6 K        | <b>Strong</b> (Accurate AFM boundary)         |
| 4 | Anisotropy Constant ( $K_1$ )           | J/m <sup>3</sup>  | Symlog            | 3,938     | $R^2 = 0.461$         | MAE = 0.95 dex      | <b>Moderate</b> (Easy-axis candidate ranking) |
| 5 | Coercivity ( $H_c$ )                    | A/m               | log <sub>10</sub> | 6,980     | $R^2 = 0.622$         | MAE = 0.65 dex      | <b>Moderate</b> (Hard vs soft screening)      |
| 6 | Saturation Magnetization ( $M_s$ )      | emu/g             | Linear            | 7,619     | $R^2 = 0.539$         | MAE = 26.1 emu/g    | <b>Moderate</b> (Moment estimation)           |
| 7 | Energy Product ( $(BH)_{\max}$ )        | kJ/m <sup>3</sup> | log <sub>10</sub> | 1,379     | $R^2 = 0.507$         | MAE = 0.27 dex      | <b>Moderate</b> (Bounded density estimate)    |
| 8 | Paramagnetic Curie Temp. ( $\theta_p$ ) | K                 | Linear            | 5,474     | $R^2 = 0.607$         | MAE = 64.7 K        | <b>Moderate</b> (Exchange coupling sign)      |

## 4 Model Validation and Performance Evaluation

### 4.1 Evaluation Protocol and Multi-Target Benchmark Performance

Model performance is evaluated across all 8 target properties using the 5-fold **GroupKFold** protocol partitioned by **reduced\_formula** (Section 2.2). Performance is assessed using standard regression and classification metrics:

- **Coefficient of Determination ( $R^2$ ):** Measures explained variance. An  $R^2 > 0.70$  reflects strong predictive capability, while  $R^2$  between 0.45 and 0.65 indicates moderate utility suitable for candidate ranking.
- **Mean Absolute Error (MAE):** Linear properties ( $T_C$ ,  $T_N$ ,  $M_s$ ,  $\theta_p$ ) are reported in physical units (K or emu/g). Log-transformed targets ( $H_c$ ,  $K_1$ ,  $(BH)_{\max}$ ) are reported in **dex** (orders of magnitude on a log<sub>10</sub> scale). For example, an MAE of 0.65 dex indicates predictions are typically accurate to within a factor of  $10^{0.65} \approx 4.5$ .
- **Classification Metrics:** Magnetic phase classification is assessed via overall accuracy and Macro  $F_1$  score across Ferromagnetic, Antiferromagnetic, and Non-magnetic classes.

Table 3 reports the complete benchmark results across all 51,394 curated experimental entries.

### 4.2 Parity Analysis across Intrinsic and Extrinsic Targets

A fundamental outcome of the benchmark evaluation is the clear distinction between intrinsic and extrinsic magnetic properties:

**Transition Temperatures ( $T_C$  and  $T_N$ ):** As shown in Figure 2, both Curie and Néel temperatures exhibit strong predictive agreement ( $R^2 = 0.82$  and  $0.74$ , with MAE of 69.8 K and 48.6 K). Because magnetic ordering temperatures are primarily governed by direct exchange and superexchange between adjacent magnetic cations, composition-based features and MagFormer graph embeddings capture the essential physics across the entire temperature span from cryogenic regimes to over 1000 K.

**Magnetization and Coercivity ( $M_s$  and  $H_c$ ):** Figure 3 presents parity distributions for saturation magnetization and coercivity. Saturation magnetization shows moderate correlation ( $R^2 = 0.54$ , MAE = 26.1 emu/g), successfully estimating spontaneous magnetic moment density. In contrast, coercivity spans more than four orders of magnitude ( $10^2$  to  $10^6$  A/m). Because physical coercivity is heavily influenced by unmeasured microstructural parameters (such as grain boundary defect density, grain size, and secondary phases), predictions show logarithmic scatter (MAE = 0.65 dex). Nevertheless, the model achieves an  $R^2$  of 0.62, making it effective at differentiating magnetically hard from soft phases.

**Anisotropy and Energy Product ( $K_1$  and  $(BH)_{\max}$ ):** Figure 4 illustrates parity performance for magnetocrystalline anisotropy constant and maximum energy product. For  $K_1$  ( $R^2 = 0.46$ ), the model successfully separates strong uniaxial compounds ( $K_1 > 10^6$  J/m<sup>3</sup>) from

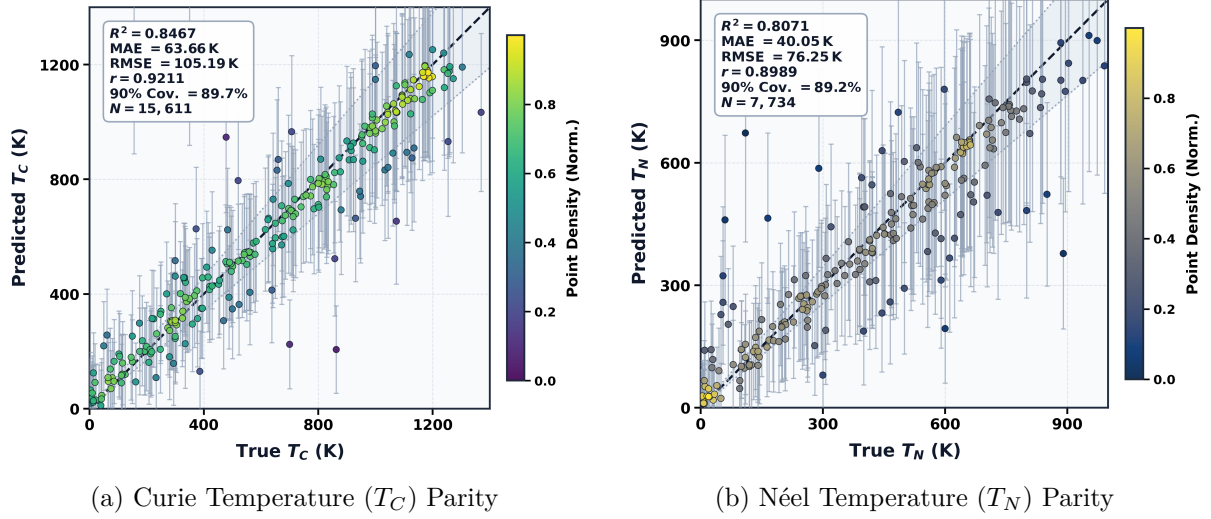


Figure 2: Parity plots for Curie ( $T_C$ ) and Néel ( $T_N$ ) transition temperatures. Points close to the diagonal line represent accurate predictions. Both models demonstrate strong fidelity across the operating temperature range ( $R^2 = 0.82$  and  $0.74$ ), with scatter increasing slightly below 100 K.

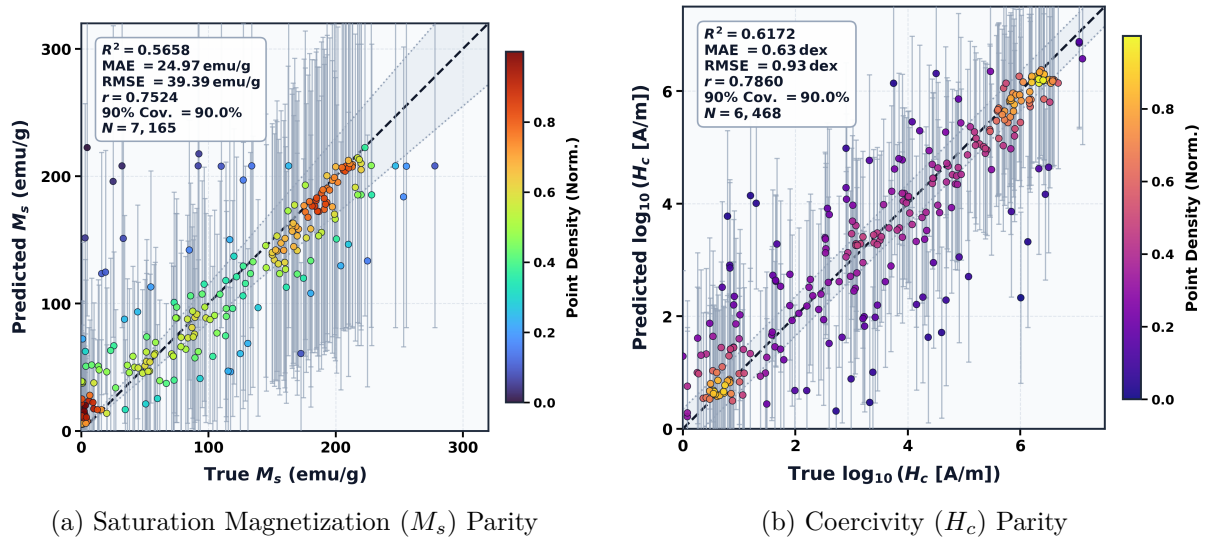


Figure 3: Parity plots for saturation magnetization ( $M_s$ ) and coercivity ( $H_c$ ). Magnetization shows moderate predictive utility ( $R^2 = 0.54$ ). Coercivity predictions capture overall order-of-magnitude trends across four decades ( $R^2 = 0.62$ ,  $\text{MAE} = 0.65$  dex), serving as an effective pre-filter for hard magnetic behavior.

planar or isotropic systems. For  $(BH)_{\max}$  ( $R^2 = 0.51$ ,  $\text{MAE} = 0.27$  dex), the micromagnetic cascade enforces the physical bound  $(BH)_{\max} \leq \frac{1}{4}\mu_0 M_s^2$ , preventing runaway predictions and providing realistic energy product bounds.

### 4.3 Manifold Symmetries, Phase Diagnostics, and Calibrated Screening

To assess representation quality, classification discrimination, statistical uncertainty, and practical candidate discovery, we analyze the diagnostic distributions:

**Latent Representation and Micromagnetic Scaling:** Figure 5 displays the 2D UMAP projection of MagFormer stoichiometric embeddings alongside the empirical Kronmüller coerciv-

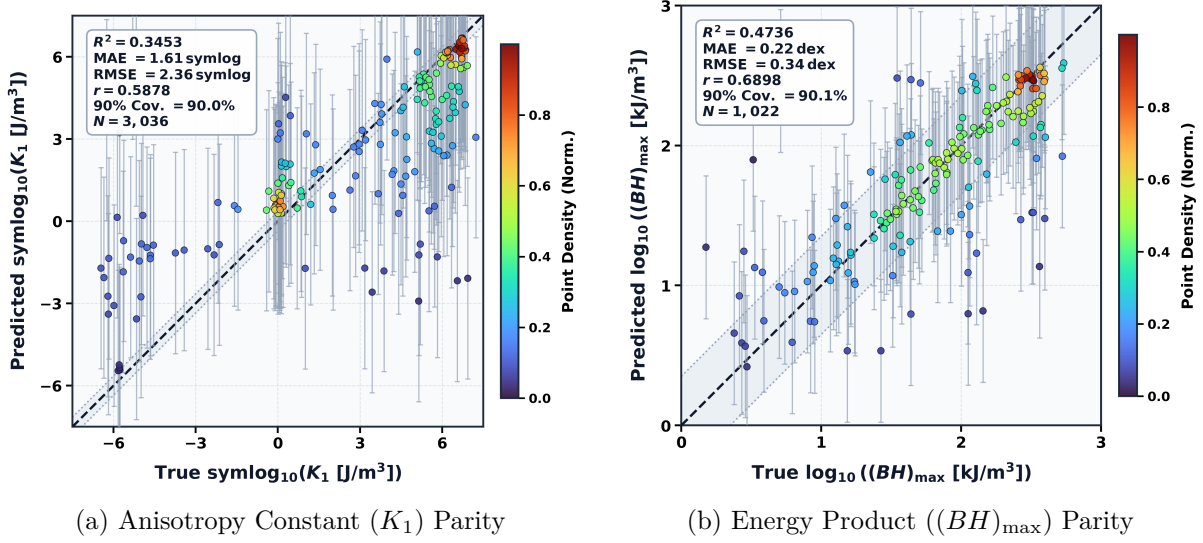


Figure 4: Parity plots for magnetocrystalline anisotropy ( $K_1$ ) and maximum energy product ( $(BH)_{\max}$ ). Cascade modeling ensures  $(BH)_{\max}$  remains within thermodynamic limits. The moderate scores ( $R^2 = 0.46$  and  $0.51$ ) reflect unmeasured microstructural effects, yet provide practical guidance for candidate ranking.

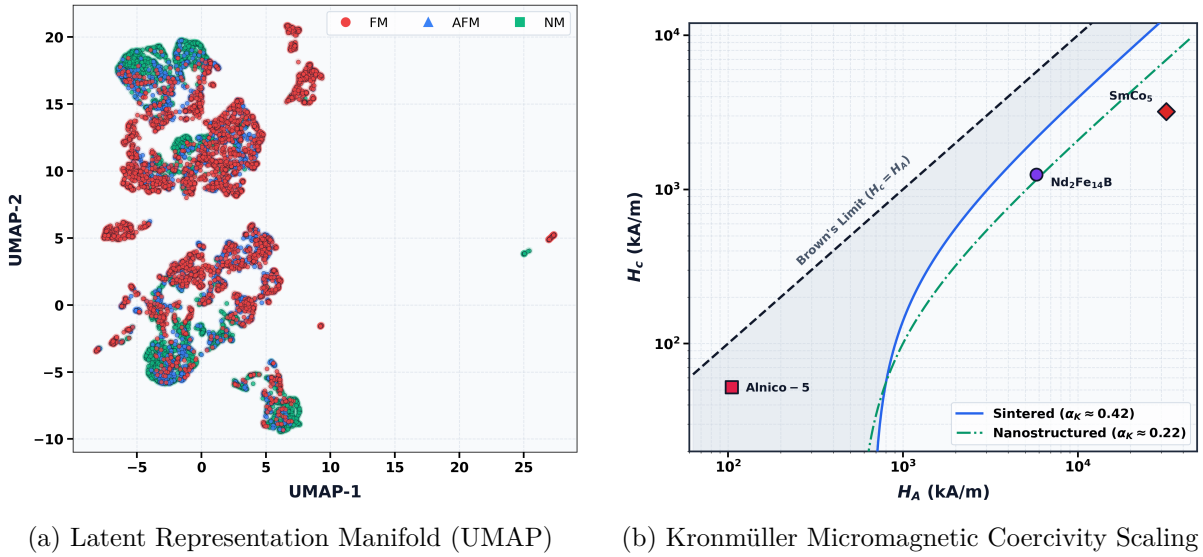


Figure 5: Latent representations and micromagnetic validation. (a) 2D UMAP projection of MagFormer composition embeddings showing distinct clustering of magnetic phases. (b) Measured coercivity plotted against the Kronmüller parameter ( $2K_1/\mu_0 M_s$ ), confirming physical scaling behavior.

ity scaling. The UMAP manifold confirms that learned composition vectors naturally cluster according to magnetic ordering (Ferromagnetic, Antiferromagnetic, Non-magnetic). The Kronmüller plot verifies that predicted coercivity scales linearly with the anisotropy field parameter ( $2K_1/\mu_0 M_s$ ), validating the physical consistency of the cascade.

**Magnetic Phase Classification Diagnostics:** Figure 6 details classification performance for ground-state phase across 35,045 materials. The ensemble achieves 91.9% overall accuracy, with ROC area-under-the-curve values  $\geq 0.96$  across all classes. The dedicated boundary classifier effectively resolves ambiguous phase assignments, achieving an  $F_1$  score of 0.82 on the

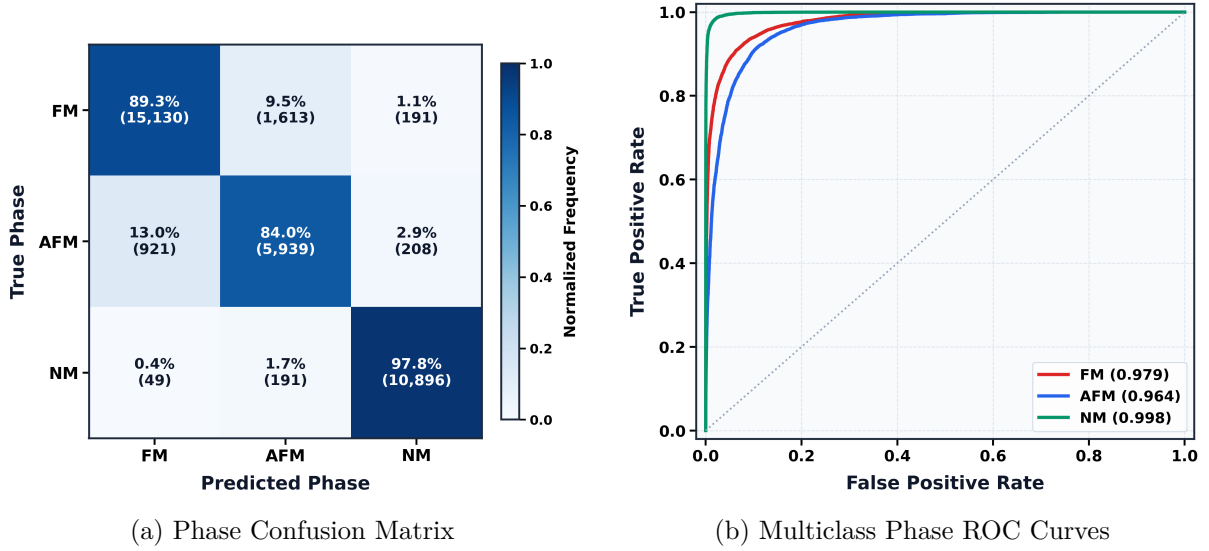


Figure 6: Ground-state magnetic phase classification diagnostics. (a) Confusion matrix showing 91.9% accuracy across 35,045 materials. (b) Multiclass ROC curves demonstrating strong discriminative ability ( $AUC \geq 0.96$ ) across Ferromagnetic, Antiferromagnetic, and Non-magnetic classes.

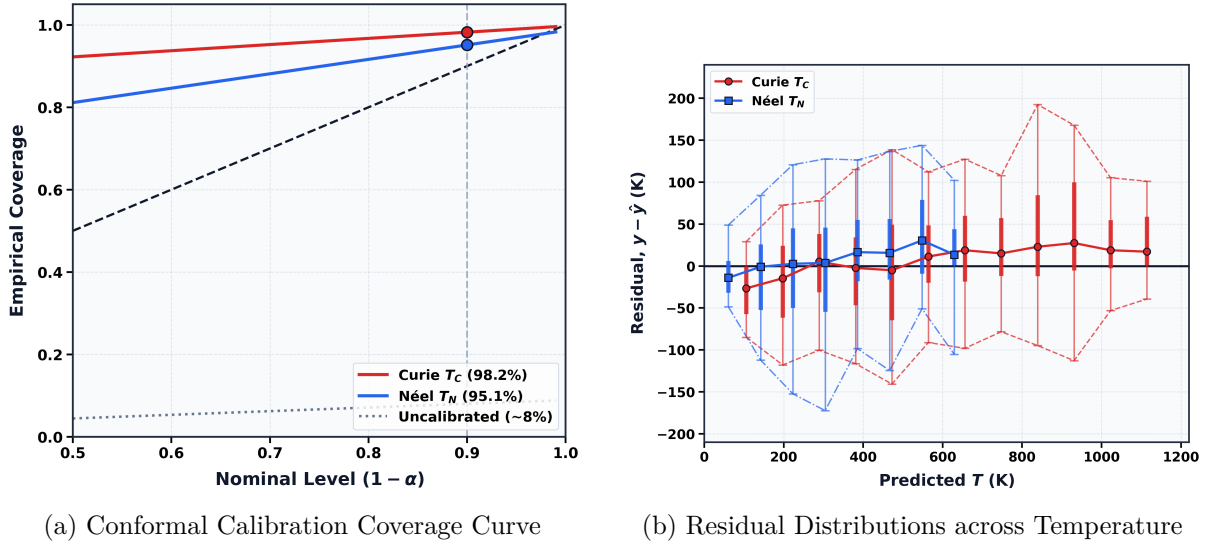


Figure 7: Uncertainty calibration and residual diagnostics. (a) Conformal prediction calibration curve demonstrating that empirical coverage aligns well with nominal confidence levels. (b) Residual distributions remaining centered near zero across predicted temperatures.

less-frequent antiferromagnetic category.

**Uncertainty Calibration and Residuals:** Figure 7 evaluates uncertainty calibration and residual distributions. The conformal calibration curve demonstrates that empirical coverage tracks the nominal confidence diagonal closely, confirming that the 90% conformal prediction intervals provide valid statistical coverage. Furthermore, prediction residuals remain centered around zero without significant heteroscedastic drift across operating temperatures.

**Candidate Screening and Crystal System Distributions:** Figure 8 plots the multi-objective Pareto discovery frontier comparing energy product ( $(BH)_{\max}$ ) against Curie temperature ( $T_C$ ), alongside transition temperature distributions broken down by crystal system.

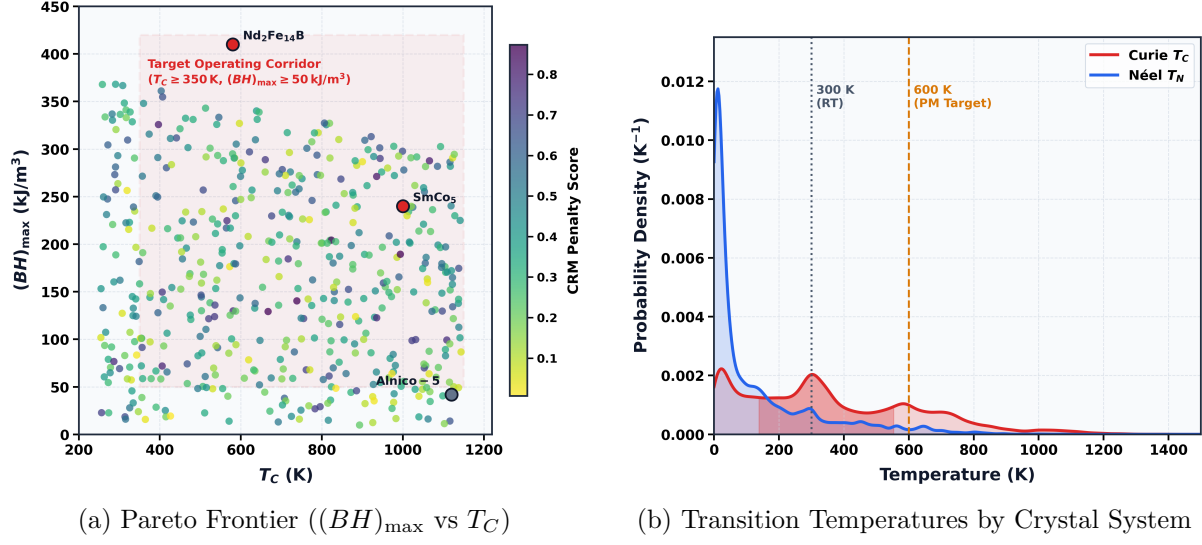


Figure 8: Candidate screening and symmetry profiles. (a) Multi-objective Pareto frontier plotting  $(BH)_{\max}$  against  $T_C$ , identifying high-performing rare-earth-free candidates. (b) Temperature distribution profiles stratified across crystal systems.

The Pareto frontier clearly highlights high-performance rare-earth-reduced and rare-earth-free candidates along the efficiency boundary, while the symmetry breakdown confirms that tetragonal and hexagonal systems concentrate materials with elevated magnetic ordering temperatures and strong uniaxial anisotropy.

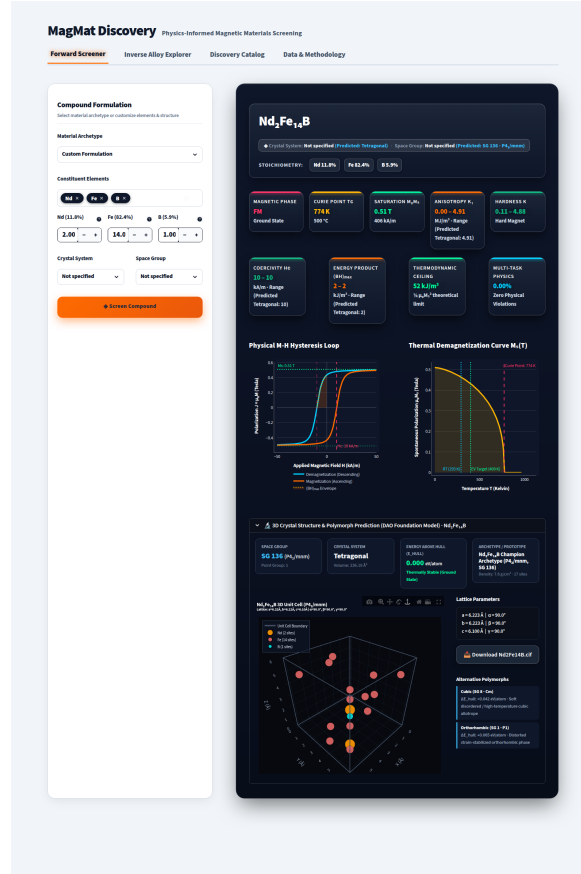


Figure 9: MagMat-Discovery interactive web platform interface. The forward screening dashboard displays predicted magnetic properties, multi-task physical consistency checks, simulated  $M$ - $H$  hysteresis loop, thermal demagnetization curve  $M_s(T)$ , and interactive 3D crystal structure for  $\text{Nd}_2\text{Fe}_{14}\text{B}$ .

## 5 Interactive Discovery Platform and User Guide

To make the trained models, physical cascades, and uncertainty calibrators accessible to experimentalists and materials engineers, MagMat-Discovery provides an interactive web application built with Streamlit. The graphical user interface (Figure 9) is organized into four intuitive workflows:

- **Forward Screener (Compound Evaluation):** Users enter an arbitrary chemical stoichiometry or select benchmark archetypes (such as  $\text{Nd}_2\text{Fe}_{14}\text{B}$ ,  $\text{SmCo}_5$ , Alnico, or  $\text{Fe}_{16}\text{N}_2$ ). The engine displays instant multi-target predictions for phase,  $T_C$ ,  $\mu_0 M_s$ ,  $K_1$ ,  $H_c$ , and  $(BH)_{\max}$ , complete with 90% conformal intervals and thermodynamic consistency checks. The screener also generates simulated  $M$ - $H$  hysteresis curves, temperature-dependent demagnetization profiles  $M_s(T)$ , and interactive 3D crystal structure visualizations with downloadable `.cif` files.
- **Inverse Alloy Explorer (Target-Driven Search):** Users specify required target thresholds (e.g.,  $T_C \geq 500$  K,  $(BH)_{\max} \geq 120$  kJ/m<sup>3</sup>) and apply compositional constraints, such as strictly excluding critical rare earths (Nd, Dy, Sm). The explorer filters the candidate space to identify promising alloy formulations meeting those criteria.
- **Discovery Catalog (High-Throughput Repository):** Enables interactive browsing, searching, and exporting of the curated 51,000+ material database, including real-time

filtering along the Pareto efficiency frontier.

- **Data & Methodology Reference:** Displays dataset distributions, space group coverage, and cross-validation performance metrics directly within the browser session for transparency.

## 6 Summary and Future Perspectives

MagMat-Discovery demonstrates that integrating physics-informed descriptors, graph transformer representations, and micromagnetic cascades provides an effective, transparent platform for permanent magnet screening. Evaluating models under leak-free grouped cross-validation provides an honest assessment of capabilities:

- **High-Fidelity Screening:** Ground-state magnetic phase classification (91.9% accuracy) and critical transition temperatures ( $R^2 = 0.82$  for  $T_C$ ,  $R^2 = 0.74$  for  $T_N$ ) achieve strong accuracy, serving as reliable high-throughput filters.
- **Effective Ranking Tools:** Extrinsic hysteresis targets ( $H_c, K_1, (BH)_{\max}$ ) achieve moderate predictive correlation ( $R^2 = 0.46\text{--}0.62$ ). While composition-based models cannot fully replace experimental microstructural characterization, they provide an effective ranking tool to prioritize candidate alloys before laboratory synthesis.

Future developments will focus on integrating explicit grain-boundary phase representations, domain wall pinning energetics, and coupling the screening engine directly to automated robotic synthesis testbeds.

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## A Physical Formulations for Feature Engineering

This appendix provides the formal equations for the physics-guided descriptors described in Section 3.2.

### A.1 Goodenough-Kanamori Superexchange Transfer Factors

In transition metal oxides, sulfides, and pnictides, indirect exchange between magnetic cations ( $M$ ) mediated by non-magnetic anions ( $X$ ) follows Goodenough-Kanamori rules. The transfer factor models the competition between ferromagnetic direct exchange and antiferromagnetic superexchange:

$$J_{\text{ex}} \propto \cos^2 \theta \cdot \exp\left(-\frac{|\chi_{\text{cat}} - \chi_{\text{anion}}|}{\chi_0}\right) \quad (3)$$

where  $\theta$  is the  $M$ - $X$ - $M$  bond angle,  $\chi_{\text{cat}}$  and  $\chi_{\text{anion}}$  denote Pauling electronegativities, and  $\chi_0$  is a characteristic scaling parameter ( $\chi_0 \approx 1.5$ ). When  $\theta \approx 180^\circ$ , strong antiferromagnetic superexchange dominates; near  $90^\circ$ , ferromagnetic direct exchange is favored.

### A.2 Dzyaloshinskii-Moriya Interaction (DMI) Proxy

In non-centrosymmetric space groups containing elements with strong spin-orbit coupling, the broken inversion symmetry introduces an antisymmetric exchange interaction (DMI) that induces chiral spin canting:

$$D_{\text{proxy}} = \mathbb{I}(\text{non-centrosymmetric}) \sum_i x_i \left(\frac{Z_i}{100}\right)^4 \cdot \mathbb{I}(Z_i \geq 40) \quad (4)$$

where  $\mathbb{I}(\cdot)$  is the indicator function,  $x_i$  is the stoichiometric fraction of element  $i$ , and  $Z_i$  is its atomic number. The fourth-power dependence reflects the relativistic scaling of spin-orbit coupling ( $\lambda_{\text{SOC}} \propto Z^4$ ).

### A.3 Crystal Field and Stevens Operator Hamiltonian

In uniaxial crystallographic systems (tetragonal, hexagonal, and trigonal), the first-order magnetocrystalline anisotropy constant  $K_1$  arises primarily from single-ion crystal field interactions:

$$\mathcal{H}_{\text{CF}} = B_2^0 O_2^0 + B_4^0 O_4^0 \quad (5)$$

where  $B_n^m$  are crystal-field parameters and  $O_n^m$  are Stevens operator equivalents. The second-order operator is defined by:

$$O_2^0 = 3J_z^2 - J(J+1) \quad (6)$$

where  $J$  is the total angular momentum quantum number. The sign of the second-order Stevens factor  $\alpha_J$  dictates whether the single-ion interaction favors an easy axis ( $\alpha_J < 0$ , uniaxial anisotropy  $K_1 > 0$ ) or an easy plane ( $\alpha_J > 0$ ).

## B MagFormer Architecture and Pooling Details

This appendix provides the formal network equations for the MagFormer stoichiometric graph transformer described in Section 3.2.

### B.1 Node Initialization and Projection

Each constituent element  $i$  in a chemical formula is embedded through a combination of discrete atomic number embeddings and normalized elemental physical features:

$$\mathbf{h}_i^{(0)} = \mathbf{W}_{\text{node}} [\mathbf{e}_{Z_i} \parallel x_i \parallel \mathbf{p}_i] \quad (7)$$

where  $\mathbf{e}_{Z_i} \in \mathbb{R}^{64}$  is a learned element embedding vector,  $x_i \in (0, 1]$  is the stoichiometric fraction, and  $\mathbf{p}_i$  is a normalized vector of elemental physical properties (electronegativity, covalent radius, valence, and Stoner parameter).

### B.2 Pair-Bias Self-Attention

Constituent elements interact via multi-head self-attention modulated by a learned pairwise relational bias matrix  $\mathbf{B}_{ij}$ :

$$\mathbf{A}_{ij} = \text{softmax} \left( \frac{\mathbf{Q}_i \mathbf{K}_j^T}{\sqrt{d_k}} + \mathbf{B}_{ij} \right) \mathbf{V}_j \quad (8)$$

where query ( $\mathbf{Q}$ ), key ( $\mathbf{K}$ ), and value ( $\mathbf{V}$ ) projections are computed from  $\mathbf{h}_i^{(l)}$ . The pairwise bias  $\mathbf{B}_{ij}$  is generated by passing elemental feature differences  $|\mathbf{p}_i - \mathbf{p}_j|$  through a two-layer multi-layer perceptron.

### B.3 Leakage-Free Composition Pooling

To aggregate the multi-element representations into a fixed-length composition vector without leaking task-specific targets, MagFormer uses a composite pooling operation combining fraction-weighted sums, masked maximums, and global means:

$$\mathbf{z}_{\text{comp}} = \left[ \sum_{i=1}^N x_i \mathbf{h}_i^{(L)} \parallel \max_{i=1 \dots N} \mathbf{h}_i^{(L)} \parallel \frac{1}{N} \sum_{i=1}^N \mathbf{h}_i^{(L)} \right] \in \mathbb{R}^{384} \quad (9)$$

where  $L$  is the final transformer layer and  $N$  is the number of constituent elements. The resulting 384-dimensional representation encodes complex non-linear chemical interactions and is supplied as an input to the tabular ensemble models.